Supplementary Information for RankCloak: Hiding Synthetic Cryptographic Artifacts in Plain English with Language Models

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ABSTRACT

This Supplementary Information documents the extended RankCloak methods and the detailed evidence supporting the revised manuscript. It specifies deterministic rank ordering, payload codecs, segmented forced spans, replay contracts, filtering, corpus construction, prompt and model identities, and work-unit accounting. It reports recovery strata, multilingual secondary tests, controlled transmission perturbations, automated readability and model-based quality diagnostics, reviewer-relevant ablation contrasts and all topic contrasts, neural detector splits and leakage sensitivity, capacity–quality relationships, computational overhead, and worked examples. Evidence classifications and unresolved boundaries are retained explicitly. The material does not constitute human evaluation, deployment validation, broad multilingual naturalness evidence, cryptographic-security proof, or a claim of robust visible-text transmission.

Supplementary Note S1: Extended RankCloak methods

Rank ordering and tie handling

For a fixed model M, token history h, vocabulary $\mathscr{V}$, next-token logit $z_M(v \mid h)$, and integer token identifier $\mathrm{id}(v)$, RankCloak orders candidates by the lexicographic key

$$\big(-z_M(v \mid h), \mathrm{id}(v)\big).$$

Ranks are one-indexed:

$$\mathrm{rank}_h(v) = 1 + \left|\{u : z_M(u \mid h) > z_M(v \mid h) \ \vee\ [z_M(u \mid h) = z_M(v \mid h) \wedge \mathrm{id}(u) < \mathrm{id}(v)]\}\right|.$$

The inverse $\mathrm{token}_h(r)$ is the unique token occupying rank r. No stochastic sampling or nondeterministic tie operation is used. The model artifact, embedded tokenizer, quantization, chat/prompt rendering, filter mask, and tie rule are therefore part of the replay configuration.

Payload codecs

Let $x = (b_1, \dots, b_m)$ be the exact serialized payload bytes.

Direct subword representation. The displayed serialization is tokenized under a declared payload-side context. If the resulting tokens are e_i, their ranks $\mathrm{rank}_{g_i}(e_i)$ form the representation. This representation can be short but its rank ceiling is the model vocabulary size; it is therefore a measurement condition rather than a bounded codec.

ASCII-byte fixed radix. For base $B \in \{8, 16\}$, set $w = \lceil \log_B 256 \rceil$. Each byte has the unique fixed-width expansion

$$b_i = \sum_{j=0}^{w-1} d_{ij} B^{w-1-j}, \qquad 0 \le d_{ij} < B,$$

and is encoded as ranks $d_{ij} + 1$. Thus ASCII $B = 8$ uses three ranks per byte, all in 1–8; ASCII $B = 16$ uses two ranks per byte, all in 1–16. Decoding subtracts one, groups exactly w digits, rejects digits or values outside the declared range, reconstructs the bytes, and verifies the expected serialized length and digest.

Hex-nibble representation. After verifying canonical lowercase hexadecimal syntax, each character $c_i \in \{0,\ldots,9,a,\ldots,f\}$ maps to its nibble value $\nu(c_i) \in [0,15]$ and rank $\nu(c_i)+1$. Decoding applies the inverse map and rejects any rank outside 1–16. This codec is not applied to base64 or hyphenated UUID display forms unless an eligible canonical hex serialization was declared.

Non-segmented encoder and decoder

Algorithm S1 Deterministic non-segmented RankCloak encoding

Require: Serialized payload x, codec C, prompt p, model/tokenizer M, filter F
Ensure: Saved cover token IDs Y, representation metadata

$R \leftarrow C.\mathrm{Enc}(x)$; $h \leftarrow \mathrm{tokenize}_M(p)$; $Y \leftarrow []$
for r in R **do**
    compute logits for $M(\cdot \mid h)$
    apply the deterministic allowed-token mask F
    order allowed tokens by decreasing logit then token ID
    $y \leftarrow \mathrm{token}_h(r)$; append y to Y and h
end for
save Y, prompt/configuration identities, codec metadata, and payload digest
return Y

Algorithm S2 Exact-copy and forced-span RankCloak decoding

Require: Saved or retokenized token IDs Y, forced-span boundaries $\mathscr{B}$, codec C, prompt schedule, model/tokenizer M, filter F
Ensure: Decoded payload $\hat{x}$ or structured failure

$\hat{R} \leftarrow []$
for each segment j in declared order **do**
    initialize h from the exact segment prompt and any declared lead-in
    for each token y within forced boundary $\mathscr{B}_j$ **do**
        recreate the filtered deterministic ordering under h
        append $\mathrm{rank}_h(y)$ to $\hat{R}$; append y to h
    end for
    ignore every token labeled natural tail
end for
$\hat{x} \leftarrow C.\mathrm{Dec}(\hat{R})$; verify length, serialization, and payload digest
return $\hat{x}$ or the first structured divergence

For a non-segmented record, $\mathscr{B}$ covers the entire generated payload sequence. For a segmented record, representation ranks $R = (r_1,\ldots,r_n)$ are partitioned into ordered half-open slices

$$R_j = R[jq : \min\{(j+1)q, n\}],$$

where q is the segment size. Forced token offsets and role masks are saved separately for each public message. Segment order and boundaries are authenticated only by the retained experimental record; RankCloak itself does not provide an authentication mechanism.

Lead-ins, natural tails, and dynamic stopping

A lead-in consists of rank-1 greedy tokens generated before a forced span. It carries no payload, but it changes every forced-token context and must be reproduced before decoding. Lead-in lengths 2, 4, 8, 16, and 32 were evaluated against the no-lead-in canonical condition.

After a segmented forced span, the dynamic policy emits greedy tokens. Stopping is permitted only after at least eight natural tokens and when the declared rules recognize terminal punctuation (`.`, `!`, or `?`) or a double newline, delimiters are balanced, and the final form is not a declared dangling connective or punctuation fragment. An emergency maximum of 256 tail tokens yields a censored stop. Fixed 20–60-token sentence completion and no-tail alternatives were also evaluated. Every tail token is labeled `tail`, carries no rank, and is excluded from forced-span decoding.

Filtering

The canonical `safe_text_filter_v1` constructs a deterministic allowed-token mask before ranking. It excludes declared unusable or non-public token forms and is applied identically at encoder and decoder. The no-filter ablation uses the complete candidate vocabulary; `roundtrip_stable_filter_v1` additionally requires isolated text/token roundtrip stability. Because removal changes rank indices, a filter is part of the key-like shared configuration rather than a cleanup step. The Mistral roundtrip-stable mask was empty under the isolated criterion, producing declared unavailable cells rather than failed trials.

Replay modes and endpoint definitions

Saved-token replay passes the original generated token IDs directly to Algorithm S2. Detokenized-text replay first renders visible text, invokes the embedded tokenizer again, and then applies declared boundary logic. Greedy-lead-in replay deterministically regenerates a declared prefix. Transformed-text replay applies exactly one frozen perturbation before tokenization. Cross-model replay uses a different declared decoder model and tokenizer.

Exact rank replay means every recovered forced rank equals the encoded rank. Exact representation recovery means the reconstructed codec representation matches. The primary endpoint is stronger: the recovered serialized bytes and their SHA-256 digest equal the source. Payload-trial, payload-group, and model–payload-group endpoints are distinct. The latter two require every constituent trial to succeed. Natural tails are never included in any rank or payload endpoint.

Supplementary Note S2: Corpus, models, prompts, and work-unit accounting

Deterministic cryptographic corpus

All 480 artifacts are public deterministic test vectors generated from declared seeds and procedures. They exercise standard formats without representing operational secrets, user credentials, or deployed traffic.

Payload class	Count	Algorithm-defined serialization
SHA-256 digest	60	Lowercase hexadecimal 256-bit digest
HMAC-SHA256 tag	60	Lowercase hexadecimal keyed-hash output under deterministic test inputs
Deterministic nonce	60	96-bit nonce serialization
128-bit token	60	Deterministically generated lowercase hexadecimal token
UUIDv4	60	Standards-conforming version/variant bits and canonical UUID display
AES-256-GCM	60	Base64 serialization of deterministic authenticated-encryption output
ChaCha20–Poly1305	60	Base64 serialization of deterministic AEAD output
Ed25519 signature	60	Base64 serialization of a deterministic signature test output

Table S1. Cryptographic test-vector corpus. Counts are balanced by class. “Real” refers to outputs of the named algorithms, not to private operational data.

Pinned model and tokenizer identities

All models used the `llama_cpp` backend, one model loaded at a time, embedded GGUF tokenizers, rank-replay batch and microbatch sizes of one, and Q4_K_M quantization. The retained manifests require exact artifact hashes and backend/library identities.

Model ID	GGUF file	Size (bytes)	SHA-256
`llama3_8b_instruct_q4_k_m`	`Meta-Llama-3-8B-Instruct.Q4_K_M.gguf`	4,920,734,272	`86c8ea6c8b755687d0b723176fcd0b2411ef80533d23e2a5030f845d13ab2db7`
`qwen2_5_7b_instruct_q4_k_m`	`Qwen2.5-7B-Instruct-Q4_K_M.gguf`	4,683,074,240	`65b8fcd92af6b4fefa935c625d1ac27ea29dcb6ee14589c55a8f115ceaaa1423`
`mistral_7b_instruct_v0_3_q4_k_m`	`Mistral-7B-Instruct-v0.3-Q4_K_M.gguf`	4,372,812,000	`1270d22c0fbb3d092fb725d4d96c457b7b687a5f5a715abe1e818da303e562b6`

Table S2. Executed model artifacts. Repository revisions for Qwen and Mistral and all upstream identities are retained in the model manifest. The three configured GGUF files are not present in the current checkout, so no fresh local generation replay was attempted for this revision.

English prompt identities

The 18 templates were fixed before the completed primary matrix. Multi-topic segmentation rotated through this schedule; single-topic segmentation reused its declared topic.

Table S3. Complete English prompt schedule.

Category	Prompt ID	Exact prompt text
Casual conversation	casual_weekend_chat	Write a friendly message to a friend about making relaxed plans for the weekend. Use ordinary conversational English and complete thoughts.
Casual conversation	casual_hobby_chat	Write a casual message to a friend describing a hobby you have been enjoying lately. Keep the tone natural, warm, and specific.
Casual conversation	casual_neighborhood_chat	Write an informal message to a neighbor about something pleasant noticed nearby. Use clear everyday language and finish the message naturally.
Professional communication	professional_project_update	Write a concise professional project update for colleagues, explaining recent progress, the next step, and one practical consideration.
Professional communication	professional_schedule_note	Write a polite workplace message proposing a meeting time and briefly explaining what the discussion should accomplish.
Professional communication	professional_process_email	Write a short professional email clarifying a routine process and inviting colleagues to raise questions. Use complete, neutral prose.
Forum Q&A	forum_practical_answer	Write a helpful answer to an online forum question about solving a common household problem. Explain the reasoning in plain English.
Forum Q&A	forum_purchase_advice	Write a balanced forum reply to someone comparing two everyday purchases. Mention useful tradeoffs without sounding promotional.
Forum Q&A	forum_learning_answer	Write a supportive question-and-answer style response to a beginner learning a practical skill. Give concrete, cautious advice.
Recipe/procedure	procedure_simple_meal	Write clear prose instructions for preparing a simple weeknight meal. Explain the order of the steps and how to tell when it is ready.
Recipe/procedure	procedure_home_task	Write an easy-to-follow explanation of a routine home maintenance task. Include preparation, the main steps, and a sensible final check.
Recipe/procedure	procedure_garden_task	Write practical instructions for a basic gardening task. Use natural sentences, explain why the steps matter, and end cleanly.
Personal narrative/blog	narrative_small_discovery	Write a short personal blog passage about discovering something useful during an ordinary day. Keep the reflection concrete and understated.
Personal narrative/blog	narrative_local_walk	Write a first-person narrative about a walk through a familiar place and one detail that changed the writer's perspective.
Personal narrative/blog	narrative_learning_moment	Write a brief personal story about learning from a minor mistake. Use coherent, natural prose and a restrained conclusion.
Factual/explanatory	explain_weather_pattern	Explain a familiar weather pattern to a general audience. Connect cause and effect clearly without using equations or specialist jargon.
Factual/explanatory	explain_everyday_system	Explain how an everyday public system works for a curious reader. Use accurate, neutral language and a logically complete explanation.
Factual/explanatory	explain_biological_process	Explain a basic biological process in accessible prose. Define the central idea, describe the sequence, and avoid unsupported claims.

Experimental matrices and reconciliation

Matrix	Declared units	Successful	Unavailable	Scope
Primary	14,400	14,400	0	Six RankCloak protocol/codec conditions plus ordinary controls
Ablation	1,872	1,824	48	Lead-in, segment size, filter, and tail policy
Robustness	3,744	3,408	336	Replay, raw transformations, limited mitigation, and cross-model mismatch
Multilingual	1,152	1,152	0	Spanish and Simplified Chinese payload/control generation
Held-out evaluator	17,280	17,232	48	Source and held-out model scoring
Detector	56	56	0	Two architectures across 28 frozen splits
Total	38,504	38,072	432	All declared work reconciled; zero execution failure; zero remaining

Table S4. Complete work-unit accounting. "Unavailable" means the declared source condition or compatible retained output did not exist; it is excluded from estimands and is not an observed experimental failure.

The 432 unavailable units comprise 48 Mistral roundtrip-stable-filter ablation units, 336 Mistral limited-canonicalization robustness units (seven conditions times 48), and 48 corresponding held-out-evaluator units. The root condition was an empty isolated roundtrip vocabulary. All other matrices completed. Primary work comprised 1,440 trials each for direct subword, ASCII $B = 8$, and ASCII $B = 16$, and 720 each for non-segmented hex, segmented single-topic hex, and segmented multi-topic hex. The detector corpus and fits are accounted as declared detector work, not as additional generation.

Supplementary Note S3: Detailed recovery and multilingual results

Stratum	Success	Total	Interpretation
Llama 3 8B, all payload protocols	2,160	2,160	Saved-token exact serialized-payload recovery
Mistral 7B, all payload protocols	2,160	2,160	Same endpoint
Qwen 2.5 7B, all payload protocols	2,160	2,160	Same endpoint
Direct subword	1,440	1,440	All three models
ASCII byte, $B = 8$	1,440	1,440	All three models; rank ceiling 8
ASCII byte, $B = 16$	1,440	1,440	All three models; rank ceiling 16
Hex nibble, non-segmented	720	720	Eligible hex-like payloads
Hex nibble, segmented single-topic	720	720	Forced spans decoded; tails ignored
Hex nibble, segmented multi-topic	720	720	Forced spans decoded; tails ignored

Table S5. Primary recovery by model and protocol. Every row uses the same exact serialized-payload endpoint. These overlapping summaries are descriptive strata, not additional independent trials.

At the frozen trial endpoint, 6,480/6,480 trials recovered, with Wilson 95% CI [0.9994075335, 1.0000000000]. At the strict payload endpoint, a payload passed only when every observed model, prompt, and protocol trial passed: 480/480, CI [0.9920605009, 1.0000000000]. At the model–payload endpoint, the result was 1,440/1,440, CI [0.9973394178, 1.0000000000]. Exact rank replay was retained as a diagnostic; the confirmatory outcome was the original serialized payload bytes and digest.

Language	Trial success	Payload-group success	Trial Wilson 95% CI	Scope
Spanish	288/288	48/48	0.986837–1.000000	Secondary saved-token exact-copy recovery
Simplified Chinese	288/288	48/48	0.986837–1.000000	Secondary saved-token exact-copy recovery
Combined	576/576	96/96	Descriptive combined total	No human naturalness or broad multilingual inference

Table S6. Secondary multilingual recovery. Each language's payload-group Wilson interval is 0.925900–1.000000.

These results do not test translation, code switching, cross-language decoding, or native-speaker judgments. They also remain conditional on the same model/tokenizer artifacts and exact saved tokens. The GGUF artifacts are absent from the current checkout; consequently the retained results can be integrity-checked against their manifests, but fresh generation or rank replay cannot be reproduced locally from this checkout alone.

Supplementary Note S4: Robustness and failure analysis

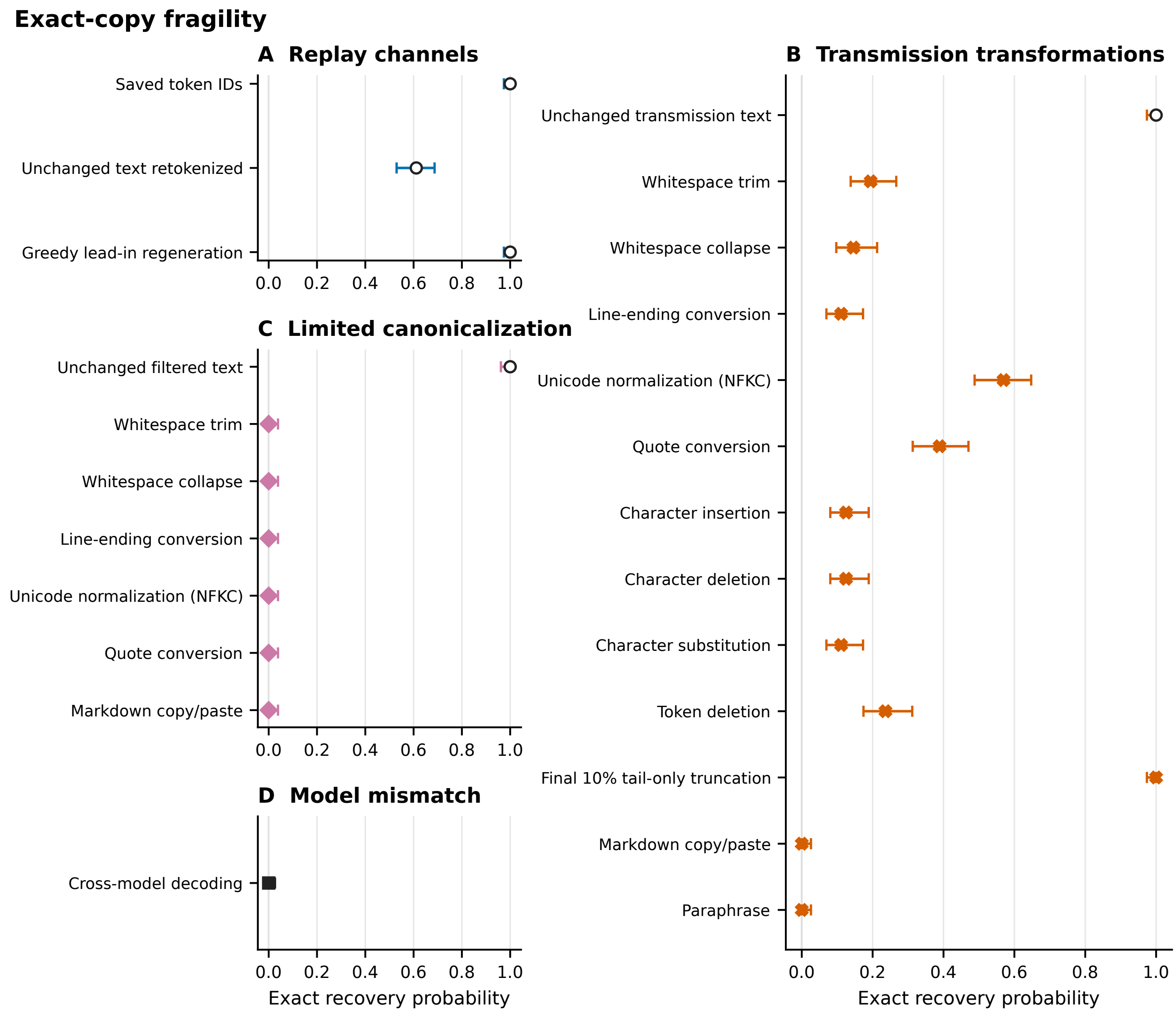

Figure S1. Complete recovery matrix and failure diagnostics. The upper panels distinguish saved-token, visible-text, transformed-text, limited canonicalization, and cross-model replay. Intervals use source-cover Wilson calculations rather than treating two cross-model decoders as independent covers. The lower panels summarize first-divergence categories. Those categories identify the first recorded mismatch and are descriptive, not causal proof. Unavailable Mistral limited-mitigation cells are excluded rather than counted as failures.

Table S7. Complete principal robustness recovery table.

Family/replay	Condition	Success	Total	Wilson 95% CI
Replay: saved IDs	Unmodified	144	144	0.974016–1.000000
Replay: detokenize/retokenize	Unmodified	88	144	0.529589–0.686859
Replay: greedy lead-in regeneration	Unmodified	144	144	0.974016–1.000000
Raw transmission	Unmodified	144	144	0.974016–1.000000
Raw transmission	Final 10% tail-only truncation	144	144	0.974016–1.000000
Raw transmission	Unicode NFKC	82	144	0.487804–0.647476
Raw transmission	Quote conversion	56	144	0.313141–0.470411
Raw transmission	Token deletion	34	144	0.174168–0.311768
Raw transmission	Whitespace trim	28	144	0.138095–0.266672
Raw transmission	Whitespace collapse	21	144	0.097405–0.212667
Raw transmission	Character deletion	18	144	0.080551–0.188937
Raw transmission	Character insertion	18	144	0.080551–0.188937
Raw transmission	Character substitution	16	144	0.069559–0.172872
Raw transmission	LF-to-CRLF line endings	16	144	0.069559–0.172872
Raw transmission	Markdown blockquote copy/paste	0	144	0.000000–0.025984
Raw transmission	Deterministic paraphrase	0	144	0.000000–0.025984
Cross-model retokenized	Unmodified, two alternate decoders	0	288 outcomes/144 covers	source-cover CI 0.000000–0.025984
Limited canonicalization	Unmodified	96	96	0.961524–1.000000
Limited canonicalization	Each of six available transformed cells	0	96 per cell	0.000000–0.038476

The limited canonicalization cells covered line endings, markdown copy/paste, quote conversion, NFKC, whitespace collapse, and whitespace trim. Each had 96 observations and 48 unavailable Mistral source conditions. The all-zero transformed results do not support a mitigation claim.

Requested class	Coverage	Frozen proxy	Boundary
Whitespace changes	Direct	Trim, collapse, line endings	Separate conditions; not a pooled estimand
Leading/trailing whitespace	Direct	Trim	Addition was not tested
Unicode normalization	Direct	NFKC	Other forms untested
Quotation marks	Direct	Straight-to-smart conversion	Reverse/locale-specific forms untested
Copy/paste	Direct	Fixed markdown blockquote pipeline	Not an application/OS survey
Line wrapping	Partial	LF-to-CRLF proxy	No visual reflow/insertion test
Markdown/email quoting	Partial	Markdown blockquote	No client headers, nesting, or reflow
Sentence boundaries	Partial	Broad paraphrase	Boundary edits not isolated
Punctuation changes	Untested	None	Alphanumeric substitution is not a punctuation test
Arbitrary prefix/suffix	Untested	None	Blockquote markup is not general insertion
Case conversion	Untested	None	No estimate available

Table S8. Perturbation-coverage inventory: five directly tested, three partially represented, and three untested requested classes.

Failure summaries classify detokenization/retokenization divergence, whitespace/markup tokenization divergence, quotation divergence, Unicode normalization divergence, content-edit divergence, semantic rewrite divergence, and model/tokenizer identity mismatch. For example, a mismatch can appear at the first forced position because a transformed prefix changes the context, even when later visible words are unchanged. The recorded first-difference position, rank-out-of-bound flag, and token/rank lengths are useful localization evidence, but they cannot distinguish all interacting causes.

Supplementary Note S5: Automated readability and model-based quality

The computational stimulus design selected 504 rows balanced over seven conditions (72 per condition) and over all 18 prompt templates. Selection and randomization metadata were frozen in the retained manifest. Metrics were computed on visible strings. Flesch Reading Ease is a syllable/sentence surface heuristic. Surface-flag counts search declared formatting and artifact patterns. TF–IDF cosine similarity measures lexical prompt overlap. None of these quantities records a reader's judgment.

Condition	Mean Flesch	Prompt-template bootstrap 95% CI	Interpretation
Ordinary control	61.940597	54.897042–68.143167	Automated control reference
Segmented full message	60.551834	56.873661–64.135677	Forced spans plus non-payload tails

Table S9. Principal automated readability rows. The complete figure includes all seven balanced conditions and associated surface diagnostics.

Source-model quality supplied 45 protocol/model contrasts; 42 had Holm-adjusted $p < 0.05$, with estimates from -5.298565 to +2.684536. The independently held-out evaluator also supplied 45 contrasts, 42 adjusted $p < 0.05$, from -6.161972 to +3.287497. Signs differed by protocol and model, so there is no universal quality ordering. The source-cover-log-probability and held-out-evaluator mixed models were singular. Their coefficient tables remain diagnostic; no fixed-effects fallback was substituted.

Forced-span metrics average only payload-bearing token positions. Full-message metrics additionally include greedy tail positions and can therefore improve as the tail becomes longer even if the forced span is unchanged. Both are reported to avoid treating tail fluency as evidence about the embedded positions.

The human-study status manifest records recruitment authorized = false, survey deployed = false, participants = 0, and ratings = 0. Planning materials are retained only to document a possible future design. They were not administered and are not reproduced here as a completed instrument. A model-name mismatch in those planning materials (Llama 3.1 versus the executed Llama 3 model) further prevents treating them as an approved executed protocol.

Supplementary Note S6: Ablations and topic effects

Locked ablation extraction

The ablation matrix paired 48 payload groups within each available model. The canonical setting was no lead-in, eight ranks per segment, the safe-text filter, and dynamic semantic stopping. Differences in Table S10 are level minus canonical. These compact contrasts were extracted after outcomes were available, without fitting new models, and are classified as exploratory/post-outcome. Payload group is the pairing and 2,000-resample bootstrap unit. The printed table presents the reviewer-relevant contrasts; the complete 60-row machine-readable contrast table, including multiplicity fields and secondary outcomes, remains in the public repository.

Across the evaluated 2-, 4-, 8-, 16-, and 32-token conditions, lead-ins generally worsened likelihood, and longer lead-ins generally increased full-message length. The 32-token contrast had mean-log-probability difference -1.652327 (95% CI, -1.749091 to -1.553103) and full-token-count difference +159.590278 (125.829861 to 191.030035). This is an adverse boundary/context result, but the paired contrast is not a causal decomposition. Segment-size changes altered efficiency and cover length: moving from 8 to 32 ranks increased effective rate by 1.231461 bits per full token (1.073116 to 1.388188) and decreased full-message length by 174.500000 tokens (-208.405035 to -140.573264). Forced-token count is codec-determined and unchanged; the length difference arises from fewer segment tails.

Removing the filter changed mean log probability by -0.001797 (-0.037738 to 0.035394) and effective rate by -0.062969 (-0.134265 to 0.007651); both selected intervals include zero. The Mistral roundtrip-stable-filter condition was unavailable for 48 work units, so the stable-filter zero rows use two models, whereas the no-filter comparison uses three. These contrasts do not establish a filter benefit.

Relative to dynamic stopping, no tail increased effective rate by 3.171551 bits per full token and reduced full-message length by 254.951389 tokens; removing non-payload text cannot be interpreted as a cover-quality improvement. The fixed 20–60-token sentence-tail policy reduced full-message length by 65.708333 tokens. Dynamic stopping checks a frozen sentence-boundary/length predicate and then applies an emergency cap; it is not a reader judgment.

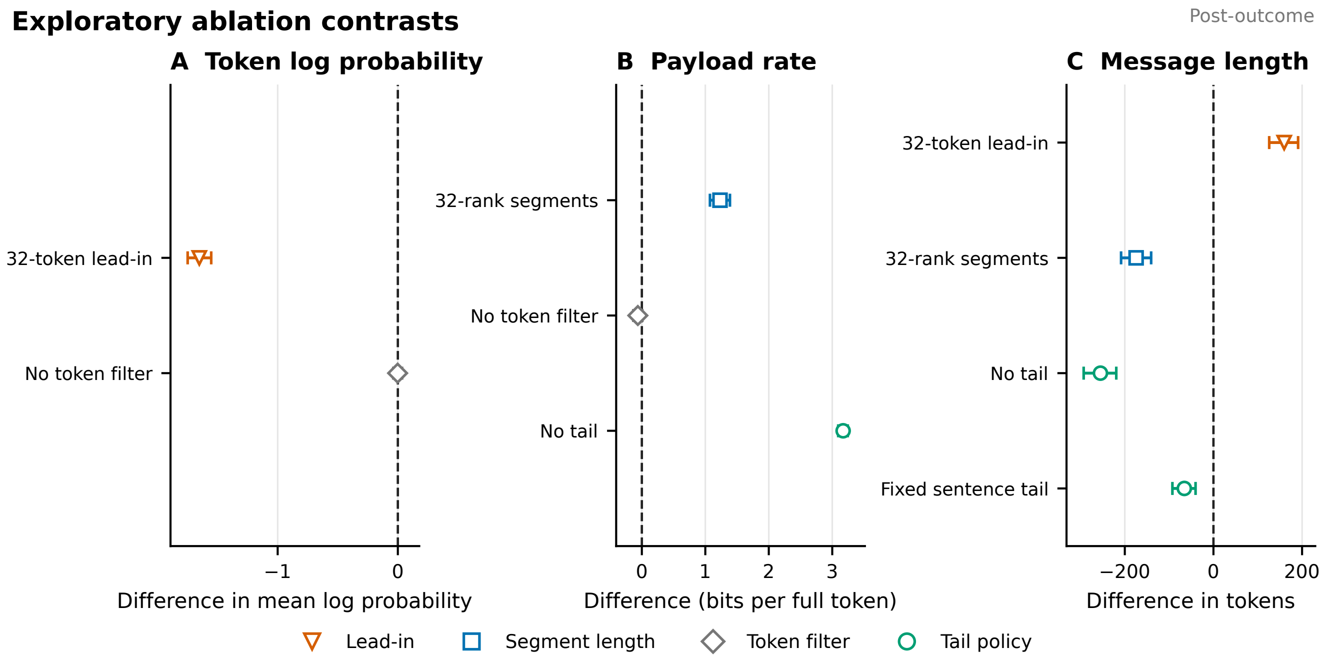

Figure S2. Ablation summary. Selected paired level-minus-canonical estimates are shown with payload-bootstrap 95% confidence intervals. Rows are exploratory/post-outcome evidence extraction from retained outputs. The Mistral roundtrip-stable-filter cell is unavailable for 48 work units; unavailable is not an execution failure. No-tail rate gains primarily reflect deletion of non-payload tokens and are not quality evidence.

Factor and level	Δ mean log probability (95% CI)	Δ full tokens (95% CI)	Δ bits/full token (95% CI)	Interpretation or availability
Lead-in: 2 tokens	-0.575880 [-0.667347, -0.491716]	+13.534722 [-21.431424, 50.179861]	-0.063224 [-0.157433, 0.024628]	Likelihood lower; length and rate intervals include zero
Lead-in: 4 tokens	-0.788830 [-0.883630, -0.694343]	+0.520833 [-36.346181, 35.365104]	-0.051850 [-0.130954, 0.029064]	Likelihood lower; length and rate intervals include zero
Lead-in: 8 tokens	-1.041886 [-1.115991, -0.961546]	+56.750000 [25.909201, 87.697396]	-0.179845 [-0.262452, -0.101276]	Lower likelihood, longer cover, lower rate
Lead-in: 16 tokens	-1.367867 [-1.450430, -1.285953]	+124.645833 [85.268229, 164.115799]	-0.299132 [-0.379449, -0.216430]	Lower likelihood, longer cover, lower rate
Lead-in: 32 tokens	-1.652327 [-1.749091, -1.553103]	+159.590278 [125.829861, 191.030035]	-0.388736 [-0.469659, -0.312450]	Largest adverse lead-in contrast
Segment size: 4 ranks	+0.116065 [0.045376, 0.192279]	+212.527778 [172.067535, 253.639236]	-0.373965 [-0.451966, -0.300845]	More segment tails; longer cover and lower rate
Segment size: 16 ranks	+0.064803 [-0.000267, 0.131660]	-104.694444 [-135.813021, -74.589410]	+0.524216 [0.418241, 0.640763]	Fewer tails; shorter cover and higher rate
Segment size: 32 ranks	+0.152306 [0.076589, 0.230426]	-174.500000 [-208.405035, -140.573264]	+1.231461 [1.073116, 1.388188]	Fewer tails; shortest selected cover contrast
No filter	-0.001797 [-0.037738, 0.035394]	+30.090278 [1.998264, 59.432986]	-0.062969 [-0.134265, 0.007651]	No clear likelihood or rate benefit
Stable filter	0.000000 [0.000000, 0.000000]	0.000000 [0.000000, 0.000000]	0.000000 [0.000000, 0.000000]	Two-model invariant; Mistral unavailable, not failed
No tail	0.000000 [0.000000, 0.000000]	-254.951389 [-292.821007, -219.249479]	+3.171551 [3.094005, 3.244380]	Mechanical deletion of non-payload tokens
Fixed sentence tail	0.000000 [0.000000, 0.000000]	-65.708333 [-92.590451, -40.372917]	-0.037921 [-0.113898, 0.033508]	Shorter than dynamic; rate interval includes zero

Table S10. Reviewer-relevant ablation contrasts. Differences are level minus the canonical setting (no lead-in, eight ranks per segment, safe-text filter, and dynamic tail). Estimates pool three available models and 48 paired payloads per model except the stable-filter row, which contains two models because the 48-unit Mistral condition was unavailable. Intervals are payload-bootstrap 95% intervals. Mean log probability is evaluated on payload-bearing forced positions; zero tail-policy differences arise because tail choice does not change those positions. All rows are exploratory/post-outcome.

Prompt-topic contrasts

The topic extraction compares the multi-topic and single-topic segmented schedules within model. Artifact counts are ratios on the response scale; the remaining estimates are multi-topic minus single-topic contrasts on their reported model scales. One of 15 rows excluded the null: the Llama expected artifact-count ratio was 1.276361 (95% CI, 1.114743–1.461409; Holm-adjusted $p = 0.003294$). Every other Holm-adjusted value was at least 0.396248. This locked extraction is exploratory and post-outcome; it supports neither a general topic advantage nor a new confirmatory hypothesis.

Table S11. All 15 prompt-topic contrasts.

Outcome	Model	Estimate	95% CI	p_{Holm}
Primary artifact count	Llama	1.276361	[1.114743, 1.461409]	0.00329358
Primary artifact count	Mistral	1.015189	[0.848773, 1.214233]	1
Primary artifact count	Qwen	0.957147	[0.781516, 1.172247]	1
Source-model log probability	Llama	-0.022140	[-0.118718, 0.074439]	1
Source-model log probability	Mistral	0.016347	[-0.080231, 0.112925]	1
Source-model log probability	Qwen	0.018813	[-0.077766, 0.115391]	1
Effective artifact rate	Llama	0.012473	[-0.075124, 0.100070]	1
Effective artifact rate	Mistral	0.006792	[-0.080805, 0.094389]	1
Effective artifact rate	Qwen	-0.015822	[-0.103419, 0.071775]	1

Outcome	Model	Estimate	95% CI	p_{Holm}
Held-out evaluator log probability	Llama	-0.012801	[-0.129427, 0.103825]	1
Held-out evaluator log probability	Mistral	0.012046	[-0.104580, 0.128671]	1
Held-out evaluator log probability	Qwen	0.024711	[-0.091915, 0.141336]	1
Payload throughput	Llama	0.018304	[-0.016147, 0.052755]	0.595439
Payload throughput	Mistral	0.026471	[-0.007981, 0.060922]	0.396248
Payload throughput	Qwen	-0.009378	[-0.043829, 0.025073]	0.595439

Supplementary Note S7: Neural steganalysis and leakage sensitivity

Corpus, architectures, and splits

The frozen corpus contains 15,840 rows: 7,920 RankCloak covers and 7,920 matched ordinary controls, grouped by 480 payloads. Payload groups, rather than rows, were assigned to train, validation, and test sets. The TextCNN-equivalent implementation lowercases and tokenizes text into a 30,000-term vocabulary, truncates or pads to 256 tokens, uses 300-dimensional learned embeddings and 100 filters at each of widths 3 and 5, applies dropout 0.5, and optimizes with Adam at learning rate 0.001. The pretrained DeBERTa-v3-base classifier uses maximum length 256, learning rate 2×10^{-5}, weight decay 0.01, and three epochs.

For each architecture, the frozen design comprises one matched split, 18 held-out-template splits, three leave-one-model splits, and six leave-one-codec splits. Thus 28 fits per architecture and 56/56 fits overall completed, producing 55,440 prediction rows. No row-level reassignment was used to improve performance.

Primary reported endpoints are ROC-AUC and balanced accuracy. Precision, Brier score, and true-positive rate at false-positive rate at most 1% are explicitly supplementary/exploratory post-freeze measures. All confidence intervals for an individual split use 2,000 payload-group bootstrap resamples. Ranges across heterogeneous held-out splits are descriptive minima and maxima, not confidence intervals. Every split-level estimate and interval for all retained metrics is supplied in machine-readable Supplementary Data Table S7.1 (`supplementary_tables/detector_split_metrics.csv`).

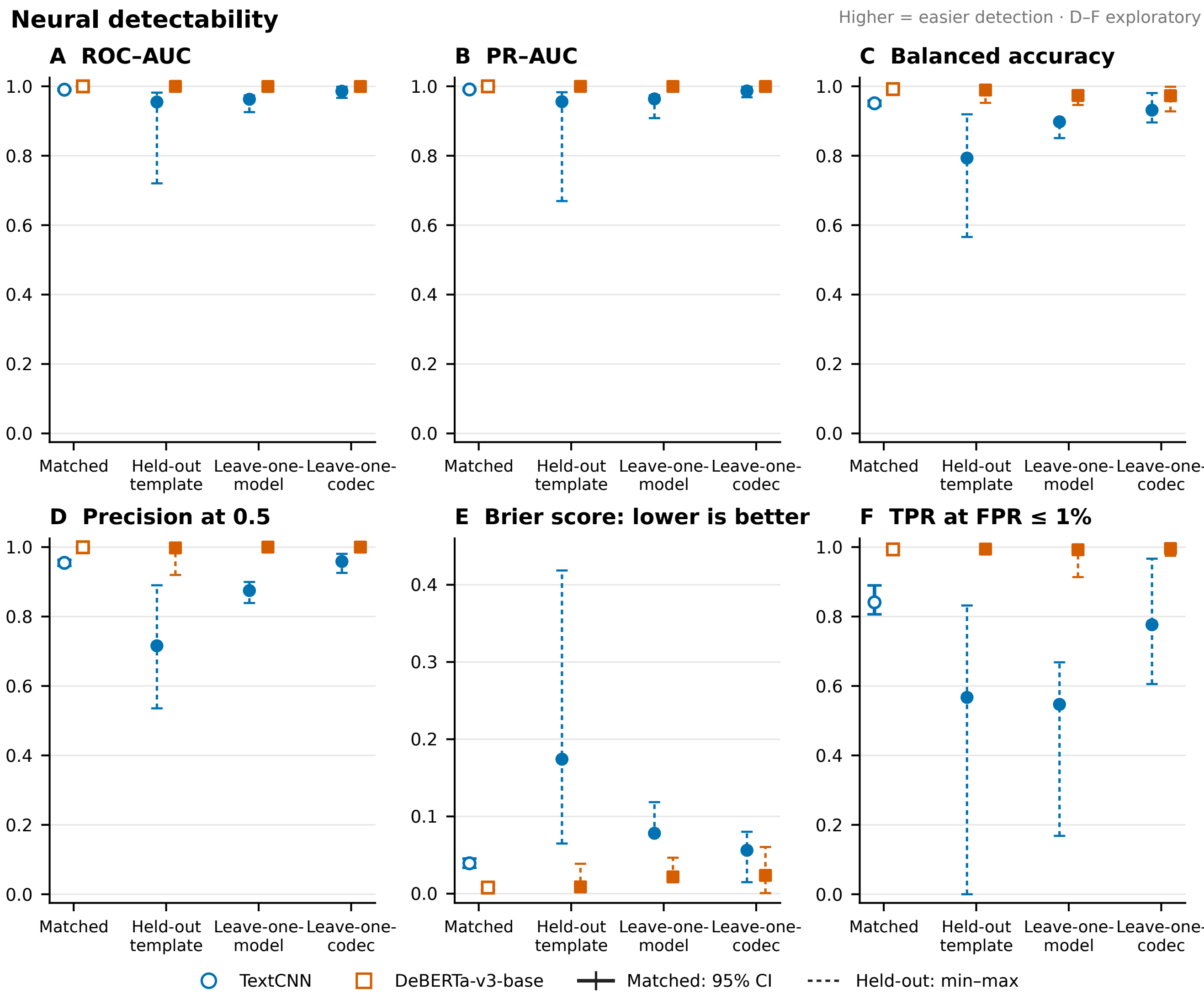

Figure S3. Full neural-detector evaluation. The six panels report confirmatory frozen endpoints together with supplementary exploratory metrics. Matched error bars are payload-group bootstrap 95% confidence intervals. Held-out regime bars span the minimum and maximum across prespecified splits and are not confidence intervals. Brier score is lower-is-better; high values in the other panels indicate stronger distinguishability and therefore weaker concealment.

Architecture	Matched metric	Estimate	Payload-bootstrap 95% CI	Status
TextCNN-equivalent	ROC-AUC	0.990190	0.987740–0.992682	Confirmatory frozen
TextCNN-equivalent	Balanced accuracy	0.951010	0.943456–0.958466	Confirmatory frozen
TextCNN-equivalent	Precision	0.954684	0.945891–0.963421	Supplementary exploratory
TextCNN-equivalent	Brier score	0.039320	0.033497–0.045453	Supplementary exploratory
TextCNN-equivalent	TPR at FPR $\leq$ 1%	0.840909	0.806517–0.889752	Supplementary exploratory
DeBERTa-v3-base	ROC-AUC	0.999834	0.999696–0.999939	Confirmatory frozen
DeBERTa-v3-base	Balanced accuracy	0.991919	0.987654–0.995665	Confirmatory frozen
DeBERTa-v3-base	Precision	0.999487	0.998393–1.000000	Supplementary exploratory
DeBERTa-v3-base	Brier score	0.007895	0.004432–0.011787	Supplementary exploratory
DeBERTa-v3-base	TPR at FPR $\leq$ 1%	0.993434	0.988724–0.998387	Supplementary exploratory

Table S12. Matched detector endpoints. Higher ROC-AUC, balanced accuracy, precision, and low-FPR sensitivity are adverse for concealment; lower Brier score indicates better detector calibration.

Architecture	Regime (split count)	ROC-AUC range	Balanced-accuracy range
TextCNN-equivalent	Held-out template (18)	0.720421–0.981705	0.565909–0.919318
TextCNN-equivalent	Leave one model (3)	0.925623–0.973650	0.850758–0.883333
TextCNN-equivalent	Leave one codec (6)	0.966532–0.998634	0.895833–0.980556
DeBERTa-v3-base	Held-out template (18)	0.996291–0.999979	0.952273–0.995455
DeBERTa-v3-base	Leave one model (3)	0.996772–0.999979	0.945455–0.970455
DeBERTa-v3-base	Leave one codec (6)	0.997917–1.000000	0.927778–0.998611

Table S13. Ranges across heterogeneous frozen held-out splits. These ranges are not pooled confidence intervals.

Leakage, exclusion sensitivity, repeatability, and cost

Exact train/test visible-text identity and exact tokenizer-sequence identity checks passed in all 28 frozen split definitions. A separate normalized character n-gram diagnostic at its declared threshold found 53 train/test near-duplicate pairs, of which 44 crossed payload groups, affecting 16 splits. Excluding affected test payload groups without retraining changed ROC-AUC from -0.001925 to +0.004568 for TextCNN-equivalent and from -0.000038 to +0.000166 for DeBERTa-v3-base; other observed endpoint shifts were also small. This threshold-dependent sensitivity does not rebuild training data and therefore does not eliminate the design limitation.

Fixed-seed reruns on the same CUDA device produced exact metric equality for both architectures. This is evidence of same-device repeatability only. CPU neural training was infeasible in the retained run and no CPU result was substituted; the study makes no CPU/GPU equivalence claim. The representative CUDA benchmark recorded TextCNN wall/training times 91.358113/9.497936 seconds and peak allocated VRAM 306,247,680 bytes; DeBERTa values were 1,536.660910/1,425.063414 seconds and 6,380,851,200 bytes.

Supplementary Note S8: Capacity–quality theory

Rank bounds and payload length

Let a radix-B codec map a payload of L bits to n_B one-indexed ranks in $\{1,\ldots,B\}$. For fixed-width digits,

$$n_B = \left\lceil \frac{L}{\log_2 B} \right\rceil$$

when the payload is treated as a bit string with an explicitly retained length; for the byte codec, the exact implementation uses $n_B = m\lceil \log_B 256\rceil$ for m bytes. Hex nibbles use $n_{16} = 2m$. Every forced position therefore selects a token whose filtered rank is at most B. Direct subword representation has no codec-imposed rank bound beyond the eligible vocabulary.

Rank-bound proposition. Assume that at each forced position the deterministic filtered ordering contains at least B eligible tokens and that encoder and replay decoder share the same ordering state. If every codec digit is in $0,\ldots,B-1$, then every selected forced token has rank in $1,\ldots,B$, and decoder inversion returns the original fixed-width digits. This follows immediately from selecting $\mathrm{token}_h(d+1)$ and replaying $\mathrm{rank}_h\{\mathrm{token}_h(d+1)\} = d+1$. The proposition is conditional on the shared state; it does not establish robustness to retokenization or context edits.

Segmentation, tails, and rate

Partition a rank sequence of length n_B into segment lengths $s_1,\ldots,s_J$, with $\sum_j s_j = n_B$. Segmentation changes the number of prompt resets and tails but not the theoretical number of payload-bearing forced positions. If segment j contains $t_j \geq 0$ non-payload tail tokens, total cover length is

$$N_{\mathrm{full}} = n_B + \sum_{j=1}^{J} t_j + \delta,$$

where $\delta \geq 0$ denotes other counted cover-extension tokens under the retained accounting convention. The positive cover-length residual is $N_{\mathrm{full}} - n_B$; it is overhead, not a recovery error. With payload length L,

$$R_{\mathrm{forced}} = \frac{L}{n_B}, \qquad R_{\mathrm{effective}} = \frac{L}{N_{\mathrm{full}}} \quad \text{bits per full token,}$$

so $R_{\mathrm{effective}} \leq R_{\mathrm{forced}}$ whenever $N_{\mathrm{full}} \geq n_B$. A no-tail policy increases the effective rate arithmetically by reducing the denominator; this relation says nothing about perceived cover quality.

Conditional probability bound

Let $p_{(1)}(h) \geq \cdots \geq p_{(B)}(h)$ be the first B filtered next-token probabilities at history h. A bounded-rank codec selects some $r \leq B$, hence

$$-\log p_{(1)}(h) \;\leq\; -\log p_{(r)}(h) \;\leq\; -\log p_{(B)}(h).$$

Averaging gives a trace-level bound on forced-token negative log probability when all three terms are retained under the same filtered distribution. It is a model-scale likelihood statement, not a theorem about grammar, semantic coherence, human naturalness, or detectability.

Empirical checks and trace boundary

The validation inventory contained 17,424 eligible retained records. Of these, 7,008 had all fields required for complete capacity/rate/quality checks. Across evaluable traces there were zero forced-position-count violations, zero rank/rate-bound violations, and zero supported quality-bound violations. The nine aggregate cells in the capacity panel of the main manuscript lay exactly on the theoretical-versus-observed forced-position identity. There were 2,976 positive cover-length residuals, attributable to natural tails or other counted cover extension.

The complete encoder–decoder cascade proposition is not directly trace-evaluable from the saved record structure: none of the retained rows simultaneously carries every intermediate object required to replay the entire proof chain as one trace. Thus the algebraic results are conditional proofs and the retained-field checks are supporting empirical validation, not a complete end-to-end trace proof. They also do not validate human naturalness.

Validation item	Result	Interpretation
Eligible records	17,424	Broad retained validation inventory
Fully evaluable records	7,008	All required trace fields present
Forced-position violations	0	Observed count matched codec requirement
Rank/rate violations	0	Evaluated bounded-codec relations held
Supported quality-bound violations	0	Model-scale trace bound held where evaluable
Positive cover-length residuals	2,976	Tail/cover overhead, not error
Complete cascade traces	0	End-to-end proposition not directly trace-evaluable

Table S14. Empirical capacity and traceability audit.

Supplementary Note S9: Computational overhead

Primary-stage timing used 12,144 matched payload rows across the 18 model–protocol cells in Table S15. Intervals are 2,000-resample payload-group bootstrap intervals. Generation time is retained inclusive wrapper wall time; representation, filter setup, model generation, and supported recovery fields are reported separately in the sealed tables but are not claimed to be perfectly isolated kernel timings. Encoding wrapper time and supported decoding measurements consequently should not be summed as independent hardware primitives.

Across the 18 cells, mean inclusive generation time ranged from 1.674368 to 16.087628 seconds and payload throughput from 13.631855 to 100.851772 bits/s; effective payload rate ranged from 0.710004 to 7.322003 bits per full token. Direct-rank and hex-nibble protocols were generally faster than the byte-expanded conditions, while segmented protocols were generally slower because they repeated prompt and tail work. These retained inclusive wall-time measurements support computational accounting under the recorded hardware and harness, not universal hardware-performance claims.

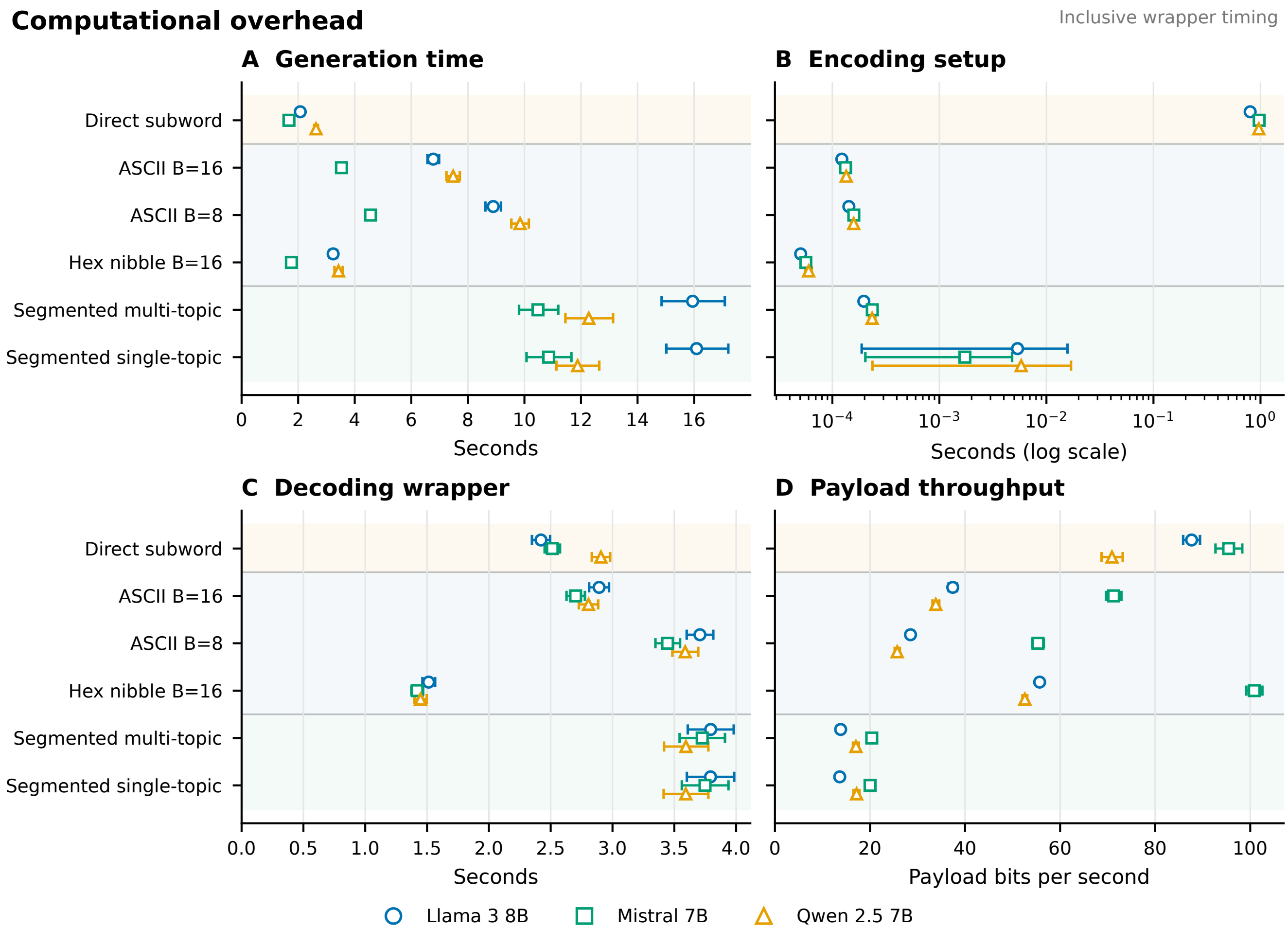

Figure S4. Complete computational-overhead summary. Primary model–protocol cells show payload-bootstrap summaries for inclusive generation, encoding setup, supported recovery, throughput, and effective rate. The setup panel uses a logarithmic axis because cells span several orders of magnitude. CPU time and repeated warm-up measurements were unavailable; wall time is not relabeled as CPU time.

Model	Protocol	n	Mean generation time, s (95% CI)	Mean payload throughput, bit/s (95% CI)
Llama	Direct ranks	480	2.074349 (2.014282–2.135219)	87.658980 (85.892805–89.433262)
	ASCII-16	480	6.776682 (6.572039–6.982913)	37.409987 (36.542507–38.322129)
	ASCII-8	480	8.896445 (8.619910–9.172562)	28.522326 (27.867278–29.213688)
	Hex-16	240	3.236699 (3.099398–3.376963)	55.691381 (55.148301–56.225745)
	Segmented multi-topic	240	15.946543 (14.850856–17.084494)	13.827671 (13.155163–14.496873)
	Segmented single-topic	240	16.087628 (15.017670–17.208756)	13.631855 (12.954794–14.325096)
Qwen	Direct ranks	480	2.632108 (2.565827–2.699101)	70.911677 (68.729360–73.171915)
	ASCII-16	480	7.480319 (7.242905–7.714463)	33.843456 (33.153343–34.571271)
	ASCII-8	480	9.847052 (9.537974–10.155572)	25.724469 (25.198423–26.278772)
	Hex-16	240	3.427555 (3.284745–3.575041)	52.584782 (52.074452–53.086870)
	Segmented multi-topic	240	12.276814 (11.449256–13.133478)	17.040206 (16.420206–17.655385)
	Segmented single-topic	240	11.881962 (11.129651–12.646529)	17.182540 (16.534412–17.783251)
Mistral	Direct ranks	480	1.674368 (1.637117–1.711610)	95.454287 (92.689931–98.318820)
	ASCII-16	480	3.532546 (3.430360–3.637370)	71.264443 (69.655729–72.892828)
	ASCII-8	480	4.558496 (4.421249–4.699172)	55.340577 (54.149134–56.561483)
	Hex-16	240	1.762009 (1.698523–1.826194)	100.851772 (99.168287–102.554774)
	Segmented multi-topic	240	10.478402 (9.810233–11.196055)	20.368846 (19.484536–21.270824)
	Segmented single-topic	240	10.858662 (10.074554–11.660671)	20.014386 (19.095928–20.969663)

Table S15. Primary model–protocol inclusive generation time and payload throughput. Llama denotes Meta-Llama-3-8B-Instruct, Qwen denotes Qwen2.5-7B-Instruct, and Mistral denotes Mistral-7B-Instruct-v0.3; all use the reported Q4_K_M configurations. Parentheses contain payload-bootstrap 95% confidence intervals.

Detector	Wall time (s)	Training time (s)	Peak allocated VRAM	Scope
TextCNN-equivalent	91.358113	9.497936	306,247,680 bytes	One fixed CUDA benchmark
DeBERTa-v3-base	1,536.660910	1,425.063414	6,380,851,200 bytes	One fixed CUDA benchmark

Table S16. Representative detector cost. CPU time, repeat warm-up measurements, and cross-device benchmarks were unavailable.

Supplementary Note S10: Worked examples and claim-boundary audit

Representative retained examples

Payload to ranks to cover. A retained Llama segmented-hex trial used the public SHA-256 test-vector serialization

`7cec7f1e68bbef4fdf066a51ccbea4377ee369e1b53219743aa386615e12d44a`.

The first eight encoded ranks were $[8, 13, 15, 13, 8, 16, 2, 15]$. Under the saved prompt/model/filter state, the associated forced token IDs were $[1789, 198, 16, 627, 13617, 10652, 25, 9842]$, detokenizing to the deliberately mechanical span

```
For\n1.\nExample conversation:  Write
```

The subsequent visible text

```
a friendly message to a friend about making relaxed plans for the weekend.
```

was a natural tail. It carried no payload rank and was ignored by the decoder. Saved-token replay reproduced all nibble ranks and the serialized digest exactly. This example illustrates deterministic rank realization, not an assertion that the forced span is human-natural.

Visible-text retokenization failure. In a retained ablation trial, detokenization followed by tokenizer reapplication first diverged in segment 6, forced position 5: expected token ID 11,623 at rank 8 became observed token ID 53,992 at replay rank 2. The visible string alone therefore did not preserve the token boundary needed by the saved-token contract.

Transformation failure. For the SHA-256 example above, straight-to-smart quote conversion failed at segment 1, position 1: expected token ID 1 at rank 9, while the transformed replay observed token ID 2,118 at rank 155. The first-divergence record localizes the mismatch but does not prove that a single changed glyph alone caused every later error.

Claim-boundary audit

Table S17. Claim boundaries applied throughout the revision.

Issue	Supported statement	Prohibited or unresolved extension
Exact-copy recovery	6,480/6,480 primary serialized payloads recovered from saved token IDs under matched replay	Ordinary visible-text reliability; cross-model robustness
Transmission	Controlled frozen perturbations reveal severe fragility; tail-only truncation preserves recovery when only ignored tokens are removed	Broad edit robustness or general mitigation
Human evaluation	504 computational stimuli have automated surface/model diagnostics	Human-perceived readability or naturalness; zero participants and ratings
Deployment	No operational channel or deployment traffic was tested	Practical covert communication or real-world validation
Detectability	Two neural architectures strongly distinguish evaluated covers; high AUC is adverse	Undetectability or resistance to future detectors
Multilingual scope	288/288 Spanish and 288/288 Simplified Chinese exact-copy secondary trials recovered	Broad multilingual naturalness or cross-language generalization
Unavailable work	432 predeclared combinations lacked compatible retained outputs	Calling unavailable combinations experimental failures
Model replay	Retained outputs are bound to model/configuration manifests	Fresh local generation replay while configured GGUF files are absent
Detector leakage	Exact identity checks passed; near-duplicate sensitivity shifts were small	Elimination of the 53-pair/16-split lexical-overlap limitation
Statistical models	Reported conditional estimates retain diagnostic warnings	Treating three singular mixed-model fits as unqualified evidence; substituting an undeclared fallback
Exploratory analyses	Compact ablation, topic, and auxiliary detector analyses are labeled post-outcome/exploratory where applicable	Recasting them as preregistered confirmatory evidence
Overhead	Retained inclusive wall-time and fixed CUDA benchmark scopes are reported	CPU time, warm-up stability, CPU/GPU equivalence, or hardware-general performance
Theory	Conditional rank/capacity relations and retained-field checks are supported	A directly trace-evaluated full cascade or human-quality proof
Cryptography	The carrier can encode outputs from declared cryptographic algorithms	Cryptographic security supplied by RankCloak itself; confidentiality without an external cryptosystem